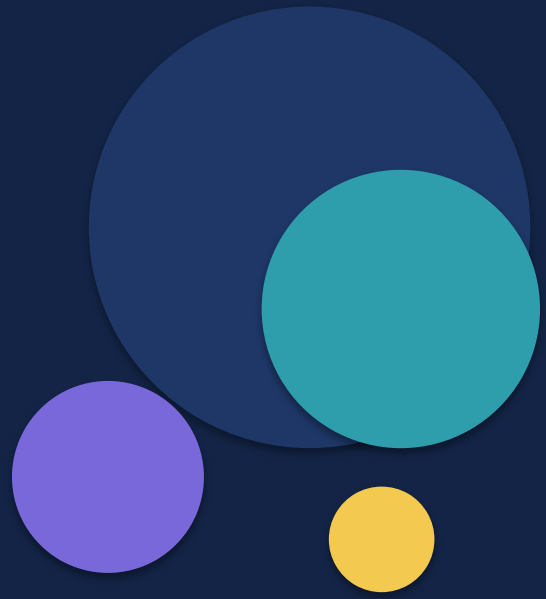
GlucoTwin India

An explainable digital twin for two-hour early warning of glucose excursions



HISTORICAL EHR + LIVE-LIKE WEARABLE STREAM → RISK ESTIMATE

Prototype uses synthetic data only. Research demonstration, not a medical device.

Zeeshan Ali Sheikh · Team DigiTech

From isolated readings to a changing patient state

The challenge is to fuse historical context with a dynamic stream and forecast one concrete outcome.

INDIA · DIABETES CONTEXT

90 million

estimated adults aged 20–79 living with diabetes in 2024

IDF Diabetes Atlas, 11th ed. (2025)

THE CARE GAP

- A lab value is a snapshot.
- Glucose and daily context change hour by hour.
- A forecast needs time-aligned signals.

OUR NARROW SCOPE

Forecast a synthetic glucose threshold crossing within two hours.

A proof-of-concept, not a diagnosis or treatment engine.

Source: IDF Diabetes Atlas (2025); WHO Diabetes Fact Sheet. Statistics are context, not model-training data.

A forecast that is specific enough to test

The model only forecasts new threshold crossings when the current simulated glucose is below the threshold.

1

Twin state

Patient profile + latest stream

2

Forecast

P(crossing ≥180 mg/dL in 2h)

3

Review

Check signal and context

4

Human decision

Clinician remains in control

180 mg/dL is a prototype label chosen for this simulator — not a universal clinical alert threshold.

Two streams make one patient-specific state

Static data explains baseline context; dynamic signals show what is changing now.

STATIC / EHR

Historical / patient context

- Age, BMI and diabetes duration
- HbA1c and hypertension flag
- Synthetic adherence factor

DYNAMIC STREAM

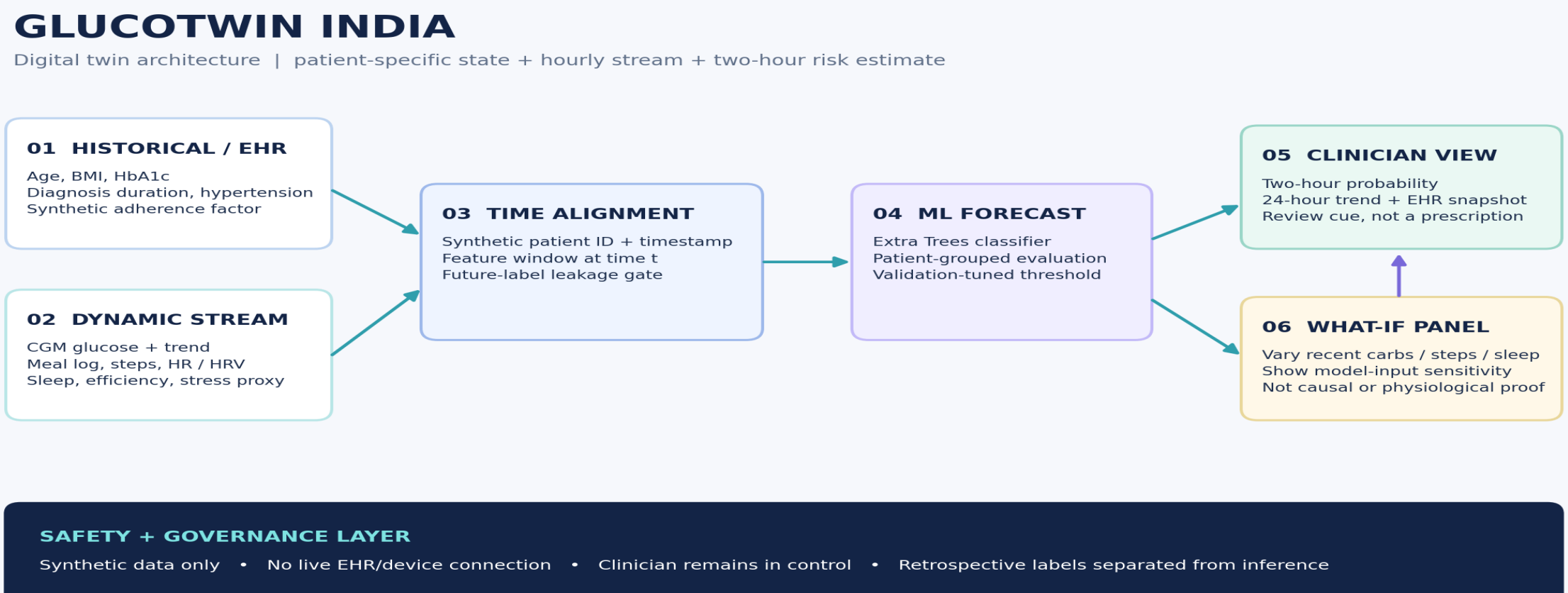
Hourly observations

- CGM value, trend and variability
- Meal carbs + steps/activity
- Heart rate, HRV and sleep
- Time since meal + stress proxy

TIME ALIGNMENT · FEATURE WINDOW AT TIME t · NO FUTURE DATA INPUTS

A transparent path from synthetic data to review

Architecture keeps inference-time features separate from retrospective outcome labels.



Research prototype. No clinical validation; do not use for diagnosis, treatment or emergency decisions.

Reproducible simulation — no patient data required

The simulator generates patient-specific trajectories and labels each timestamp from the following two hours.

What is generated?

- 240 synthetic patient profiles
- 14-day timeline; last 2 hours are label context
- 80,160 rows total across the full stream
- Meals, activity, HR/HRV, sleep and CGM-like values

Target construction

At time t, use only current and past features.

Positive label: either t+1h or t+2h reaches ≥180 mg/dL.

Rows already ≥180 mg/dL are excluded from the early-warning target and surfaced separately.

Future glucose is used only to create the retrospective label. It is not included in FEATURE_COLUMNS.

Patient-level evaluation to reduce leakage

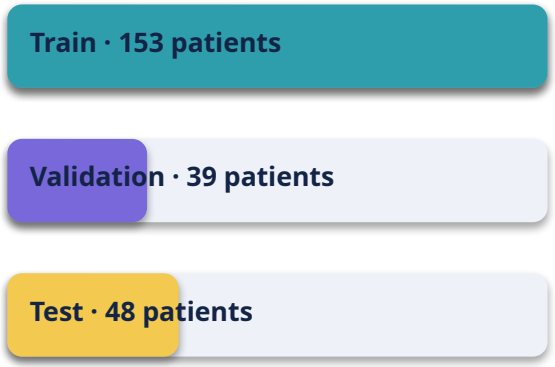
An Extra Trees baseline estimates probability; the operating threshold is tuned on validation data only.

MODEL

Extra Trees

- Nonlinear tabular baseline
- Validation threshold tuned for F2
- Global feature importance

SPLIT BY PATIENT ID



THRESHOLD

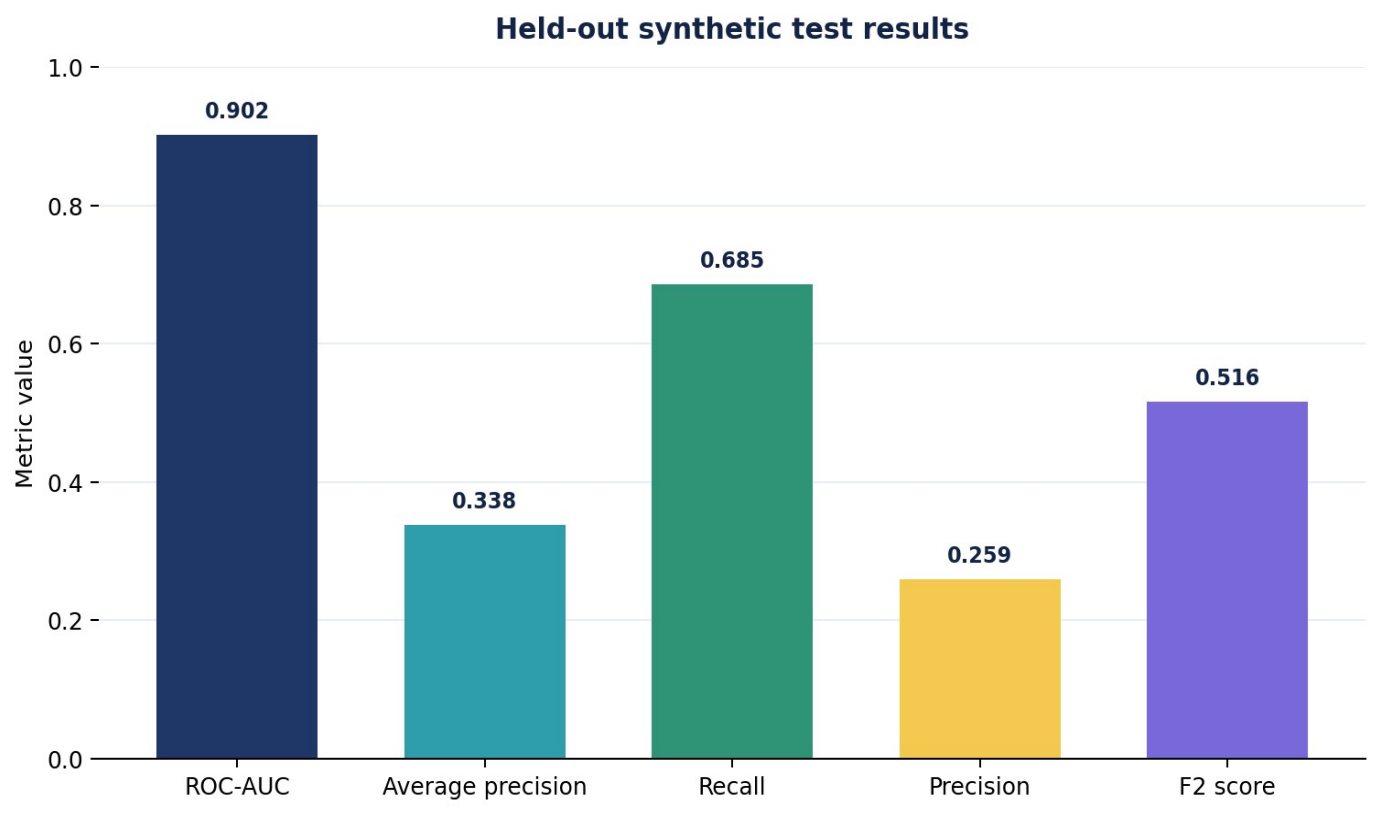
F2 tuned

Validation threshold chosen to emphasise recall for an early-warning demo; false alerts remain a trade-off.

No patient can appear in more than one split. Metrics are estimates on a synthetic domain only.

Held-out synthetic test results

These numbers are computed by the included training script; they are not clinical performance claims.



AT THE SELECTED THRESHOLD

68.5%

recall

25.9%

precision

High recall comes with false alerts.

Test set: 48 held-out patients, 15,184 eligible hourly rows. Positive event prevalence ≈ 4.5% overall.

A clinician-facing, patient-specific dashboard

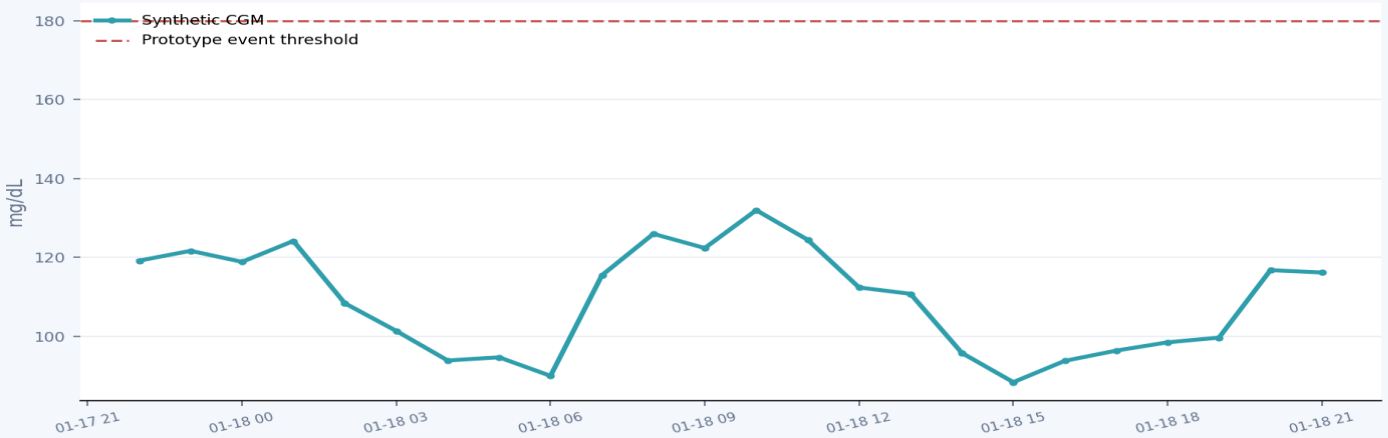
The UI connects the probability to a 24-hour trend, EHR context and a safe review workflow.

GlucoTwin India

Patient-specific glucose excursion early warning | UI preview for demo

CURRENT CGM	2-HOUR MODEL RISK	HISTORICAL CONTEXT	WEARABLE SIGNAL
116 mg/dL	**0%**	**HbA1c 5.9%**	**82 bpm**
18 Jan 2026 · 21:00	Synthetic-domain estimate	BMI 30.5 · 1y duration	HRV 56 ms

Glucose trajectory · last 24 hours



FORECAST & REVIEW CUE

Below model review threshold

Estimated chance of 180 mg/dL crossing in 2h: 0%

Review recent CGM signal quality, meal/activity context and history.

Clinician review only — no prescriptions.

HISTORICAL EHR SNAPSHOT

AGE	BMI	DIABETES DURATION	HYPERTENSION	ADHERENCE
35	30.5	0.8 y	Yes	51%

All data shown are synthetic. The UI preview is rendered from the prototype data/model; it is not a real clinical record.

Visual preview rendered from the included synthetic data and trained model. The repository includes the working Streamlit application.

Let reviewers explore sensitivity without claiming causality

The what-if panel changes model inputs; it does not simulate full human physiology or prove that an intervention causes a safer outcome.

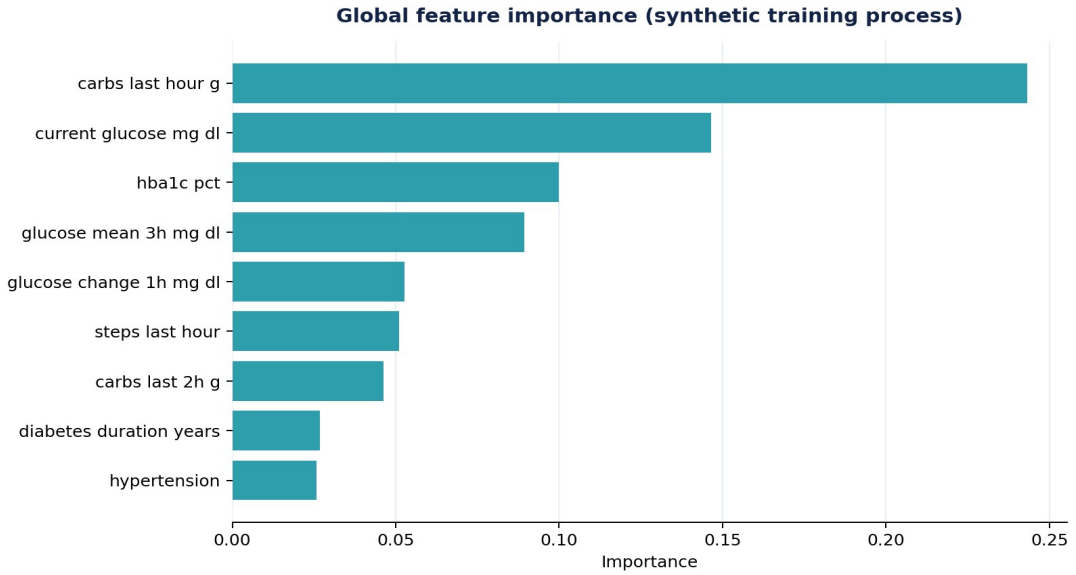
WHAT-IF INPUTS

Carbs last hour

Steps last hour

Sleep last night

Forecast updates when the scenario is applied.



Global feature importance ≠ causality or a local explanation.

Guardrails are part of the design

The prototype is deliberately positioned as a research demo and clinician review aid, not an automated decision maker.

WHAT IT DOES

- Joins two synthetic data streams
- Estimates one narrowly scoped event
- Shows metrics and data lineage

WHAT IT DOES NOT DO

- Use real patient records or devices
- Diagnose or recommend medication
- Prove clinical efficacy or causality

WHAT COMES NEXT

- External, governed datasets
- Calibration + subgroup evaluation
- Clinician workflow and safety review

Synthetic data only · Not validated for real-world use · Never use to make diagnosis, treatment or emergency decisions.

A small digital twin — end to end

GlucoTwin India demonstrates the complete challenge loop: patient state → hourly stream → forecast → clinician review.

01

FUSE

Historical + dynamic data

02

FORECAST

Two-hour event probability

03

EXPLAIN

Model results + limits

04

KEEP HUMAN

Clinician makes decisions

Next milestone: evaluate on governed real-world data with clinical partners before any real deployment.

Thank you · DigiTech · GlucoTwin India

Context sources and reproducibility

Claims about public-health context are sourced; all model data and evaluation in this repo are synthetic.

International Diabetes Federation — Diabetes Atlas, 11th Edition (2025)

https://diabetesatlas.org/media/uploads/sites/3/2025/10/IDF_Diabetes_Atlas_11th_Edition_2025_WEB.pdf

India estimate: 90.0 million adults aged 20–79 living with diabetes in 2024.

World Health Organization — Diabetes Fact Sheet

https://www.who.int/news-room/fact-sheets/detail/diabetes

Overview of diabetes, complications, detection and management.

Digital Twin Challenge 2026 — Unstop / Happiest Health

https://unstop.com/hackathons/crp-digital-twin-challenge-2026-happiest-health-1757873

Challenge scope, two data streams and repository/video submission checklist.

Implementation artefacts, generated metrics and limitations are available in the public repository README and docs.